QFT-Inspired Spectral Edge Filtering for Hybrid Visual Information Systems*

Seonmin Kim and Sang Min Yoon†
College of Computer Science , Kookmin University, Seoul, South Korea
{k990216,smyoon}@kookmin.ac.kr

Abstract

E-business information systems increasingly process document-style visual assets, including scanned forms, advertisements, reports, specifications, file folders, and news documents. Edge detection is a low-level but important operation for extracting layout boundaries, text-block structures, and other visual cues before higher-level indexing, classification, or analysis. This paper studies a quantum Fourier transform (QFT)-inspired spectral edge-filtering framework as an exploratory quantum-image-processing primitive for future hybrid quantum-classical visual information systems. Rather than claiming quantum speedup, the study analyzes how separable Sobel and Prewitt filters can be represented as one-dimensional row-wise and column-wise QFT-domain operations. Each row or column signal and its kernel are converted into amplitude-compatible quantum states through zero extension, power-of-two embedding, and normalization. QFT-domain statevectors are multiplied element-wise, re-normalized, reloaded into a circuit, and reconstructed through inverse QFT. The implementation is therefore explicitly positioned as a statevector-level quantum-circuit-inspired pipeline rather than a hardware-native quantum filtering circuit. Experiments are performed on ten class-wise RVL-CDIP document samples resized to the same 64×64 input resolution for both classical and QFT-based filtering. The zero-padded linear-aligned QFT variant preserves thresholded edge support very closely, with Edge-IoU of 0.9999 for Sobel and 1.0000 for Prewitt, while MSE, PSNR, and SSIM reveal remaining intensity-level discrepancies. Runtime results show that the current QFT statevector implementation is substantially slower than classical separable filtering, reinforcing the interpretation of the method as an analytical step toward hybrid quantum visual information processing rather than a demonstrated quantum-advantage algorithm.

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† Corresponding author

1 Introduction

Edge detection is a fundamental operation in image processing because boundaries, contours, and intensity transitions provide structural information for subsequent recognition, indexing, verification, and analysis stages [1]. In e-business information systems, visual data are not limited to product photographs or scanned receipts. Enterprise platforms also process forms, advertisements, reports, specifications, folders, invoices, and other document-style visual assets. In these settings, edge information can support document layout analysis, text-block localization, form boundary extraction, document image indexing, and visual quality assessment.

Classical gradient-based operators such as Sobel and Prewitt remain widely used because they are simple, interpretable, and computationally inexpensive for small kernels [2], [3]. However, the continuing evolution of information systems motivates the study of alternative computing models, including hybrid quantum-classical processing and quantum-inspired visual analytics. The purpose of this work is not to replace mature classical document-analysis pipelines. Instead, it examines how a familiar visual primitive can be expressed and evaluated within a QFT-based spectral circuit framework.

A transform-domain view is natural because convolution or correlation can be reformulated through spectral-domain multiplication followed by an inverse transform [1]. The quantum Fourier transform is the quantum counterpart of the discrete Fourier transform and is a central primitive in quantum algorithms [4], [5]. This paper investigates a statevector-level QFT-domain formulation in which one-dimensional image signals and one-dimensional kernels are encoded into amplitude-compatible quantum states, transformed by QFT, combined in the spectral domain, and reconstructed by inverse QFT.

The contributions are as follows. First, we connect QFT-based edge filtering to document-style visual information processing in e-business information systems. Second, we formulate a separable QFT-domain filtering pipeline for Sobel and Prewitt operators while clarifying the difference between circular spectral filtering and zero-padded linear-aligned QFT filtering, hereafter denoted as QFT-linear. Third, we report resource and runtime behavior to avoid unsupported efficiency or quantum-advantage claims. Fourth, we evaluate the method on class-wise RVL-CDIP document samples using MSE, PSNR, SSIM, Edge-IoU, Edge-F1, and runtime.

2 Related Works

The Sobel and Prewitt operators are classical discrete differentiation filters for estimating local intensity gradients [2], [3]. Their continued use is due to their interpretability, small kernel size, and simple implementation. Both operators are separable, allowing a two-dimensional filter to be implemented through two successive one-dimensional filters. This property reduces the cost of direct spatial filtering and also provides a convenient bridge to row-wise and column-wise spectral processing [1]. The QFT is one of the central building blocks of quantum computation and can be implemented using Hadamard gates, controlled phase rotations, and qubit-reversal operations [4], [5], [12]. Quantum image processing has explored image representation, transformation, and manipulation using quantum states [7], [8]. Representative image encodings such as FRQI and NEQR provide general quantum-state models for storing pixel or grayscale information [9], [10]. The present work is narrower: it does not propose a general quantum image representation, but studies how separable edge filtering can be expressed as a QFT-domain spectral operation over amplitude-encoded one-dimensional signals. E-Business Information Systems increasingly depend on visual information, including scanned business documents, identity and transaction records, product images, logistics labels, industrial inspection imagery, and service-related image streams.

Edge filtering is a low-level but important primitive in these pipelines because it supports layout segmentation, contour extraction, defect boundary localization, and structure-preserving preprocessing. A QFT-based formulation of such a primitive is relevant to EBISION when it is interpreted as an exploratory component of future hybrid quantum-classical visual analytics rather than as a standalone computer vision speedup claim. The security and privacy implications of this hybrid implementation also require careful interpretation. The present implementation assumes a trusted hybrid computing environment. Since the QFT-domain product vector is extracted, normalized classically, and reloaded into a new circuit, the current method does not provide end-to-end privacy preservation. For security-sensitive e-business settings, practical deployment would require additional mechanisms such as trusted execution, secure hardware isolation, blind quantum computation, encrypted-domain processing, or carefully designed quantum-safe document authentication protocols. These directions are beyond the current implementation but motivate the relevance of analyzing where classical-quantum handoff points occur.

3 Methods

3.1 Overview

Let $I \in \mathbb{R}^{H\times W}$ denote a document image. Instead of flattening the full image into a single vector, the proposed method exploits the separability of Sobel and Prewitt filters and processes the image through one-dimensional stages. For a row or column signal $x \in \mathbb{R}^n$ and a one-dimensional kernel $k \in \mathbb{R}^m$, the QFT-inspired spectral filtering stage is conceptually written as

$$y = \mathrm{IQFT}(\mathrm{QFT}(x) \odot \mathrm{QFT}(k)),$$

where $\odot$ denotes element-wise multiplication in the spectral domain.

In the actual implementation, x and k are first zero-padded, embedded into power-of-two-length amplitude vectors, and normalized before QFT is applied.

3.2 Separable Sobel and Prewitt filtering

The proposed framework is evaluated using Sobel and Prewitt edge filters because both are classical gradient operators and can be decomposed into one-dimensional derivative and smoothing components. The horizontal and vertical Sobel operators are written as

$$K_{S_x} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad K_{S_y} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

They can be factorized as

$$K_{S_x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} [-1 \quad 0 \quad 1], \qquad K_{S_y} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} [1 \quad 2 \quad 1].$$

Similarly, the Prewitt operators are

$$K_{P_x} = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix},\ K_{P_y} = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

They are decomposed as

$$K_{P_x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} [-1 \quad 0 \quad 1], \qquad K_{P_y} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} [1 \quad 1 \quad 1].$$

This separability allows a two-dimensional edge filter to be implemented as successive one-dimensional row-wise and column-wise operations. Let $\mathcal{Q}_k^{row}(\cdot)$ denote the proposed QFT-linear one-dimensional filtering operation applied independently to each row with kernel k, and let $\mathcal{Q}_k^{col}(\cdot)$ denote the same operation applied independently to each column. For Sobel filtering, using the derivative vector $d = [-1\ ,0\ ,1]$ and the smoothing vector $s_S = [1, 2, 1]$, the horizontal and vertical gradient responses are computed as

$$G_x^S = \mathcal{Q}_{s_S}^{col}\left(\mathcal{Q}_d^{row}(I)\right), \qquad G_y^S = \mathcal{Q}_{s_S}^{row}\left(\mathcal{Q}_d^{col}(I)\right).$$

For Prewitt filtering, the same derivative vector $d = [-1, 0, 1]$ is used with the smoothing vector $s_P = [1, 1, 1]$:

$$G_x^P = \mathcal{Q}_{s_P}^{col}\left(\mathcal{Q}_d^{row}(I)\right), \qquad G_y^P = \mathcal{Q}_{s_P}^{row}\left(\mathcal{Q}_d^{col}(I)\right).$$

The final edge-response maps are then computed as gradient magnitudes:

$$E^S = \sqrt{(G_x^S)^2 + (G_y^S)^2},\ E^P = \sqrt{(G_x^P)^2 + (G_y^P)^2}.$$

Thus, the image is not flattened into a single global vector. Instead, QFT is used as a one-dimensional spectral filtering primitive inside a separable two-dimensional edge-filtering pipeline.

3.3 Correlation, convolution, and boundary alignment

A key implementation detail is the distinction between correlation, convolution, and finite-length spectral multiplication. The classical baseline in this paper uses same-padding correlation, which is common in image-processing implementations of Sobel and Prewitt filters. In correlation, the kernel is applied without flipping, whereas convolution applies a reversed kernel.

Finite Fourier-domain multiplication naturally produces circular behavior over the chosen transform length. Therefore, a direct spectral product may not match the same-padding linear correlation used by the classical reference. To align the QFT-domain operation with the classical baseline, this paper uses a zero-padded linear-aligned QFT variant, denoted as QFT-linear. For a one-dimensional signal of length n and a kernel of length m, the transform length is chosen as

$$L = 2^{\lceil \log_2(n+m-1) \rceil}.$$

The signal is zero-padded to length L, and the kernel is embedded into the same transform length. Although the classical reference is correlation, the spectral implementation matches it by using the equivalent convolution form with a reversed kernel. After inverse QFT, the corresponding n-length same-region response is cropped to match the same-padding classical reference. This alignment ensures that the classical and QFT-based edge maps have the same visible input and output resolution. In the

RVL-CDIP experiment, both pipelines operate on the same 64×64 resized document images, and QFT padding is used only internally for one-dimensional state preparation and reconstruction.

3.4 Preprocessing before quantum encoding

For each row or column signal $x \in \mathbb{R}^n$ and one-dimensional kernel $k \in \mathbb{R}^m$, the vectors must be converted into amplitude-compatible quantum states. Using the transform length $L = 2^{\lceil \log_2(n+m-1) \rceil}$, the signal and kernel are embedded into L-dimensional vectors $\tilde{x}$ and $\tilde{k}$. The embedded vectors are then normalized as

$$|x\rangle = \frac{\tilde{x}}{\|\tilde{x}\|_2}, \qquad |k\rangle = \frac{\tilde{k}}{\|\tilde{k}\|_2}.$$

The original norms $\|\tilde{x}\|_2$ and $\|\tilde{k}\|_2$ are retained for scale restoration after inverse QFT. If either norm is zero, the corresponding filtering response is set to zero. This preprocessing step is necessary because a valid amplitude-encoded quantum state must have unit norm and length 2^q, where $q = \log_2 L$ is the number of qubits used for the one-dimensional signal.

3.5 QFT-domain filtering and scale restoration

After amplitude encoding, QFT is applied to the signal and kernel states:

$$|X\rangle = QFT|x\rangle, \ |K\rangle = QFT|k\rangle.$$

At the statevector level, spectral filtering is implemented by component-wise multiplication:

$$F = X \odot K.$$

Since F is generally not a unit-norm quantum state, it is normalized before inverse QFT:

$$|\hat{F}\rangle = \frac{F}{\|F\|_2}.$$

The reconstructed spatial response is obtained by

$$\hat{y}_{norm} = IQFT(|\hat{F}\rangle).$$

Finally, the original signal and kernel scales are restored as

$$\hat{y} = \hat{y}_{norm} \cdot \|\tilde{x}\|_2 \cdot \|\tilde{k}\|_2 \cdot \|F\|_2.$$

The real-valued part of $\hat{y}$ is used as the one-dimensional filtering response, and the response is cropped back to the original signal length n. This procedure is repeated along rows and columns according to the separable Sobel or Prewitt decomposition. Because the current implementation requires statevector extraction, element-wise multiplication, re-normalization, and circuit re-initialization, it should be interpreted as a QFT-inspired statevector pipeline rather than a hardware-native quantum filtering circuit.

3.6 Implementation model and resource implications

The present pipeline consists of amplitude preparation, QFT, statevector extraction, spectral multiplication, product normalization, re-initialization, and IQFT. The QFT circuit for q qubits has a compact gate structure compared with a full dense unitary transform, but the overall pipeline is dominated by data loading, statevector-level manipulation, and reloading in the current simulation setting. Therefore, the method should not be described as computationally faster than classical separable filtering unless runtime and resource data support such a claim.

The implementation is analyzed relative to two main viewpoints used in the final RVL-CDIP experiment: direct classical separable filtering and the zero-padded linear-aligned QFT statevector pipeline. Circular QFT filtering is discussed only as a boundary-mismatch case because finite-length spectral multiplication naturally induces circular behavior when zero-padding and cropping are not handled consistently. The final experiment reports the linear-aligned variant because it is the most appropriate counterpart to the same-padding classical correlation baseline.

Table 1: Resource and runtime comparison on RVL-CDIP 10-class samples at 64×64 input resolution. Runtimes are reported as mean ± standard deviation per image.

Method	**Main operations**	***Expected cost driver***	**Hardware realism**	**Runtime**
Classical separable filtering	Row/column same-padding correlation	*O(HWm), m = 3*	Direct classical baseline	Sobel: 0.101±0.020 s Prewitt: 0.104 ± 0.022 s
QFT zero-padded linear-aligned filtering	State preparation, QFT, spectral product, reloading, IQFT	*State preparation and statevector manipulation*	Simulation-level	Sobel: 10.748±0.751 s Prewitt: 10.355 ± 1.435 s

4 Experiments

4.1 Dataset and sampling protocol

Experiments are conducted on RVL-CDIP document-style visual samples. Ten classes are used in the reported visualization and metric table: letter, form, email, handwritten document, advertisement, scientific report, scientific publication, specification, file folder, and news article. One representative image is selected from each class using a fixed streaming and sampling procedure. Each selected document image is converted to grayscale and resized to 64×64 before filtering. Both the classical and QFT-based pipelines operate on the same resized input resolution. QFT uses power-of-two padding only internally for one-dimensional state preparation and reconstruction; the final edge maps are cropped back to the same 64×64 size before visualization and metric computation.

4.2 Evaluation metrics

The classical separable edge map is used as the reference, and the QFT-based edge map is evaluated against it. Thus, the reported metrics do not compare the original document image with an edge map. They compare classical Sobel with QFT Sobel and classical Prewitt with QFT Prewitt under the same input resolution. MSE and PSNR measure pixel-level intensity discrepancy, SSIM measures structural similarity in the edge-response maps, and Edge-IoU and Edge-F1 measure binary edge-support agreement after thresholding. Edge-F1 is computed with a one-pixel tolerance to account for small boundary localization differences. Runtime is measured separately for classical separable filtering and QFT-based filtering.

4.3 Baselines and ablations

The baseline is direct classical separable Sobel or Prewitt filtering with same-padding correlation. The QFT result uses the zero-padded linear-aligned variant because it is the most appropriate comparison to the classical same-padding reference. Circular QFT filtering is discussed as a source of boundary mismatch, but the final RVL-CDIP experiment reports the linear-aligned variant.

5 Results and Analysis

5.1 Quantitative results

Table 2 summarizes the quantitative comparison between classical filtering and QFT-based filtering on the ten RVL-CDIP class-wise samples. Under the selected thresholding protocol, the QFT-linear variant achieves nearly identical binary edge support, with Edge-IoU of 0.9999 for Sobel and 1.0000 for Prewitt, and Edge-F1 of 1.0000 for both filters. However, MSE, PSNR, and SSIM show that the grayscale edge-response intensity is not reproduced exactly. This supports the interpretation that the current QFT-inspired pipeline is contour-support preserving but not pixel-exact.

Table 2: Classical versus QFT-linear edge-map similarity on RVL-CDIP 10-class samples at 64×64 input resolution. Values are mean ± standard deviation over ten class-wise samples.

Filter	QFT variant	MSE	PSNR (dB)	SSIM	Edge-IoU	Edge-F1
Sobel	linear	0.077±0.028	11.34±1.44	0.275±0.278	0.9999±0.0003	1.000±0.000
Prewitt	linear	0.086±0.031	10.87±1.43	0.272±0.278	1.0000±0.0000	1.000±0.000

5.2 Qualitative results

Figures 1 and 2 show class-wise RVL-CDIP examples. Each figure contains ten columns corresponding to the ten document classes. The first row shows the original resized document images, the second row shows classical separable filtering, and the third row shows QFT-based filtering under the same 64×64 input resolution. The QFT maps retain major page boundaries, text-block regions, advertisement structures, and rectangular layout cues, but local brightness and edge thickness differ from the classical maps. These visual observations are consistent with Table 2: binary edge support is preserved, while intensity-level metrics remain moderate.

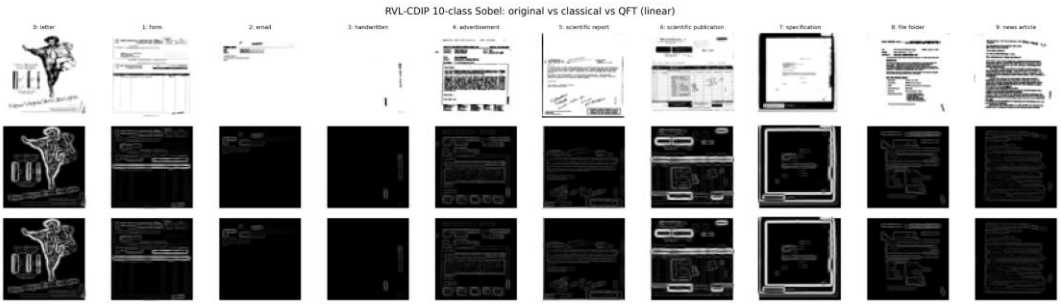


Figure 1: Class-wise RVL-CDIP Sobel edge-filtering results under the same 64×64 input resolution. The first row shows original document images, the second row shows classical separable Sobel filtering, and the third row shows QFT-linear Sobel filtering.

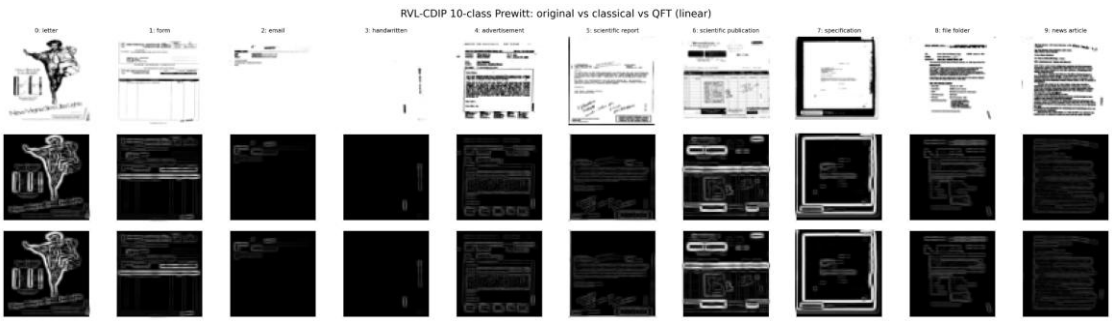


Figure 2: Class-wise RVL-CDIP Prewitt edge-filtering results under the same 64×64 input resolution. The first row shows original document images, the second row shows classical separable Prewitt filtering, and the third row shows QFT-linear Prewitt filtering.

5.3 Interpretation

The RVL-CDIP result is relevant to the scope of e-business information systems because the evaluated samples are document-style visual assets, including forms, advertisements, reports, specifications, file folders, and news documents, rather than generic benchmark images. Nevertheless, the experiment should not be interpreted as document understanding, OCR, or document classification. The goal is narrower: to examine whether a QFT-inspired quantum image-processing primitive can preserve layout-level edge support in document visual data. The low-to-moderate PSNR and SSIM values indicate that the QFT pipeline does not reconstruct the classical grayscale edge response exactly. The high Edge-IoU and Edge-F1 values indicate that, after thresholding, coarse edge support is strongly aligned with the classical reference.

Runtime results further clarify the implementation status. On 64×64 inputs, classical filtering takes roughly 0.10 s per image, while QFT statevector filtering takes roughly 10-11 s per image. The present implementation therefore does not demonstrate computational efficiency. Instead, it identifies the computational bottlenecks of amplitude preparation, statevector access, re-normalization, and circuit reloading, which are central considerations for future hybrid quantum-classical visual information systems.

6 Conclusion

This paper presented a QFT-inspired spectral edge-filtering framework for document-style visual information processing in e-business information systems. By exploiting the separability of Sobel and Prewitt filters, edge detection was expressed as successive one-dimensional QFT-domain operations over amplitude-encoded row and column signals.

Classical and QFT-based edge maps were computed from the same 64×64 input resolution. The QFT-linear variant preserved thresholded edge support very closely, but it did not exactly reproduce the grayscale intensity distribution of the classical edge maps. The method should therefore be understood as a contour-support-preserving quantum-circuit-inspired approximation rather than as a pixel-exact edge detector.

The runtime analysis also shows that the current statevector implementation is slower than classical separable filtering. This confirms that the present work does not demonstrate quantum speedup or hardware-native quantum image filtering. Its contribution is analytical: it clarifies how a classical edge primitive can be mapped into a QFT-domain hybrid pipeline, where approximation gaps arise, and why state preparation and statevector manipulation remain major bottlenecks. Future work should evaluate

larger real-world document and product-image datasets, investigate hardware-realistic implementations of spectral multiplication, and study privacy-preserving or secure hybrid architectures for sensitive business visual information.

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